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## Luxury vinyl plank stair noses and other moldings

### Abstract

A molding is made from a first flooring plank. A first groove is formed into the first flooring plank with a first flat bottom surface. A second groove is formed into the first flooring plank with a second flat bottom surface. The first flooring plank is folded at the first groove and second groove. A second flooring plank is disposed adjacent to the molding. A color and pattern of the first flooring plank and second flooring plank match.

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## Background/Summary

CLAIM OF DOMESTIC PRIORITY (1) The present application is a continuation-in-part of U.S. patent application Ser. No. 17/336,925, now U.S. Pat. No. 11,873,645, filed Jun. 2, 2021, which claims the benefit of U.S. Provisional Application No. 63/034,204, filed Jun. 3, 2020, which application is incorporated herein by reference.

FIELD OF THE INVENTION
(1) The present invention relates in general to stair noses and other moldings made from luxury vinyl plank flooring, and to methods, tools, and machines for forming the stair noses and other moldings from luxury vinyl plank flooring.
BACKGROUND OF THE INVENTION
(2) Flooring manufacturers and installers have tried many different methods for providing custom stair noses that match the surrounding floor. Typical methods involve cutting off the existing stair nose and then installing a replacement stair nose closely matching the floor being installed, as shown in FIGS. **1***a*-**1***d*. FIG. **1***a* illustrates a stair step **10** with a tread **12** and riser **14**. Nose **16** is a part of tread **12** delineated with a dotted line to show where the nose will be cut off. FIG. **1***b* illustrates a floor plank **20** with a replacement nose **22** being installed over tread **12**. Nose **22** is pretty similar to nose **16** that was already part of the underlying tread **12**, but is designed to reasonably match the color and pattern of flooring being installed in an adjacent room.
(3) FIG. **1***c* illustrates continuing to install flooring planks **30** next to nose plank **20**. FIG. **1***d* shows plank **30** installed. Additional flooring planks **30** will continue to be installed next to each other to

fully cover the stair tread or perhaps an entire room in the case of the top step.
(4) One problem that occurs with replacing stair treads along with the rest of the adjacent flooring is matching the wood grain pattern and color. Even when the exact same type of wood and finish is used for both nose plank **20** and flooring plank **30**, the color and pattern are usually off. All the floor planks **30** being used are usually made together at the same factory at the same time to match practically exactly. However, nose planks **20** are typically formed separately and, while they may match flooring planks **30** closely, will almost always have a noticeable difference in color and pattern due to being manufactured at a different time or even a different factory.
(5) Luxury vinyl plank (LVP) flooring is a modern type of flooring that is susceptible to the problems of color matching stair nosing and other molding. FIG. **2***a* shows a cross-sectional view of one plank **50** of LVP flooring. LVP flooring is typically formed of a core **52**, padding **54**, an image layer **56**, and a clearcoat finish **58** formed over the image layer. Core **52** is commonly a stone polymer or wood-plastic composite. A stone polymer core **52** is composed of calcium carbonate (limestone), polyvinyl chloride (PVC), and optionally plasticizers. Wood-plastic composite cores are similarly composed, with the addition of a wood product, such as sawdust or wood flour. A foaming agent may be added to soften the floor made with planks **50**.
(6) The desired design for the flooring is printed on image layer **56** and then attached to core **52**. Image layer **56** can be a vinyl sheet or another printable substrate. Clearcoat layer **58** typically consists of anywhere from 1 to 100 layers of clearcoat or more. Usually between 10 and 25 layers of clearcoat are used. Clearcoat layer **58** protects the printed image layer **56**, and plank **50** as a whole, from wear.
(7) Luxury vinyl plank flooring is typically formed with connectors **60** around the perimeter of planks **50** so that individual planks can be clicked or snapped together with other adjacent planks to easily form a floor with proper alignment and a seamless transition between planks. FIG. **2***a* shows a connector **60***a* on one side of plank **50** and a connector **60***b* on the other side. When two pieces of LVP flooring **50***a* and **50***b* are slid together as shown in FIGS. **2***b* and **2***c*, connector **60***a* of one plank and connector **60***b* of the other plank slide into each other. A detent is commonly used to snap the connectors together, maintaining alignment and eliminating visible gaps between planks.
(8) While LVP flooring makes installing a beautiful floor easier, LVP does not eliminate the problems of matching stair nosing to the surrounding flooring. The closest matching hardwood nosing is usually used even though the vinyl planks are printed. Achieving an exact match is very difficult. Therefore, a need exists for an improved stair nose, as well as other types of molding, that matches LVP flooring planks.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1***a*-**1***d* illustrate replacing a stair nose as part of installing flooring;
(2) FIGS. **2***a*-**2***c* illustrate luxury vinyl plank flooring;
(3) FIGS. **3***a*-**3***i* illustrate cutting and folding a luxury vinyl plank to form a stair nose;
(4) FIGS. **4***a*-**4***c* illustrate an alternative groove cut profile;
(5) FIGS. **5***a*-**5***e* illustrate saw blades used to cut grooves into the luxury vinyl plank for folding;
(6) FIGS. **6***a* and **6***b* illustrate saw blades used to cut the alternative groove profile;
(7) FIGS. **7***a*-**7***c* illustrate table saw configurations used to cut grooves into luxury vinyl planks;
(8) FIGS. **8***a*-**8***c* illustrate a table setup to fold and glue the luxury vinyl planks;
(9) FIGS. **9***a*-**9***d* illustrate forming T molding out of a luxury vinyl plank;
(10) FIGS. **10***a*-**10***e* illustrate forming end molding out of a luxury vinyl plank;
(11) FIGS. **11***a*-**11***g* illustrate forming a stair tread from a larger plank; and
(12) FIGS. **12***a*-**12***h* illustrate forming a stair tread from thinner vinyl flooring.

## DETAILED DESCRIPTION OF THE DRAWINGS

(13) The present invention is described in one or more embodiments in the following description with reference to the figures, in which like numerals represent the same or similar elements. While the invention is described in terms of the best mode for achieving the invention's objectives, it will be appreciated by those skilled in the art that it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the appended claims and their equivalents as supported by the following disclosure and drawings.

(14) One solution to providing stair noses that match luxury vinyl plank (LVP) flooring is to make the stair noses out of the same LVP planks that are being installed for the flooring. Using the same planks for both stair noses and the rest of the flooring means that the stair nose planks are manufactured at the same plant and under the same conditions as the rest of the flooring planks. The issues in the prior art with slight variations in manufacturing conditions resulting in slightly off colors and patterns are eliminated because stair nosing and floor planks are manufactured together.

(15) Making a stair nose out of LVP flooring involves cutting grooves into a floor plank and then folding the plank at the grooves into a stair nose shape. FIGS. **3***a* and **3***b* illustrate a floor plank **100** with a bottom surface **102** and top surface **104**. Plank **100** includes two long edges **106** and two short edges **108**. Plank **100** has a width extending from one long edge **106** to the other, a length extending between the two short edges **108** parallel to the long edges, and a thickness extending between top surface **104** and bottom surface **102**. Planks **100** have a latching mechanism built into the edges as connectors **60***a* and **60***b*, so the opposing long edges **106** are designed to interface with each other and the opposing short edges **108** are also designed to interface with each other. A connector **60** of a short edge **108** could interface with a connector of a long edge **106** if the installer wanted to get creative. Connectors **60** are optional, and some LVP flooring planks simply have flat surfaces that are designed to contact each other when installed without latching or otherwise interfacing with each other.

(16) Two grooves **110***a* and **110***b* are formed into bottom surface **102**, but not completely through plank **100** to top surface **104**. FIG. **3***c* shows additional detail of grooves **110**. Grooves **110** allow plank **100** to be folded 90-degrees along each of the grooves, thus turning the plank into a stair nose. Grooves **110***a* and **110***b* are substantially identical to create two 90-degree angles along the length of plank **100**. Grooves **110** each include a horizontal surface **112**, two vertical surfaces **114**, and two diagonal surfaces **116**. Diagonal surfaces **116** connect bottom surface **102** of plank **100** to the two vertical surfaces **114**. Vertical surfaces **114** connect diagonal surfaces **116** to horizontal surface **112**. Horizontal surface **112** is the deepest part of grooves **110** and connects the two vertical surfaces **114** to each other. Horizontal surface **112** is considered the bottom of groove **110** due to being the deepest part of the cut.

(17) In the illustrated embodiment, plank **100** is 8 millimeters (mm) thick, and groove **110** is formed to a depth of 7 and ⅓ mm, leaving a thin flat flexible portion **120** between horizontal surface **112** and top surface **104** with a thickness of ⅔ mm. A thickness of ½ mm is left as flexible portion **120** in other embodiments. The depth of groove **110** can be formed as close to image layer **56** as possible without damaging the image layer. Ideally core **52** would be completely removed but doing so without damaging printed layer **56** can be a challenge. Accordingly, a thin portion of core **52** is typically left under horizontal surface **112** by design. Core material **52** is flexible enough that a thin layer remaining still allows plank **100** to be folded at groove **110**. In one embodiment, groove **110** is formed to leave a fixed thickness of plank **100** in flexible portion **120** so that the remaining thickness of core **52** will depend on the total thickness of image layer **56** and clearcoat layers **58**.

(18) The width of horizontal surface **112**, and therefore the width of flexible portion **120** and the distance between vertical surfaces **114**, is 3.2 mm. Vertical surfaces **114** have a height of 1.6 mm, and diagonal surfaces **116** each extends off at a 45-degree angle from a respective vertical surface to bottom surface **102**. At bottom surface **102**, diagonal surfaces **114** are approximately 0.5772 inches or 14.66 mm apart. Horizontal surface **112**, vertical surfaces **114**, and diagonal surfaces **116**

all extend in the same profile shape for the entire length of plank **100**. Any of the above measurements can be customized as needed for different plank types, compositions, sizes, etc. to ensure that diagonal surfaces **116** make proper contact when folded.

(19) Grooves **110** with a flat horizontal surface **112** at the bottom of the grooves leaves a flat flexible portion **120** of plank **100** between horizontal surface **112** and the plank's top surface **104**. Flexible portion **120** has a relatively uniform thickness for a significant width, which allows plank **100** to bend uniformly along the entire width of horizontal surface **112** when the plank is folded. Diagonal surfaces **116** could meet at a point at the bottom of the groove, but bending of plank **100** would occur over a much thinner area of plank **100** and risk tearing of image layer **56**. For planks that are not as flexible, horizontal surface **112** can be made wider, allowing the plank to bend across a wider arc, or portion **120** can be made thinner to flex easier.

(20) Each diagonal surface **116** is at a 45-degree angle so that the angle between the two diagonal surfaces is 90 degrees. When plank **100** is bent across groove **110**, diagonal surfaces **116** contact each other when the plank is flexed to the same angle as exists between the diagonal surfaces. For a 90-degree bend in plank **100**, diagonal surfaces **116** should make a 90-degree angle when formed. A non-symmetrical groove could be formed with, e.g., one diagonal surface **116** at a 30-degree angle and the other at a 60-degree angle, and the diagonal surfaces would still meet when plank **100** is bent to 90 degrees. Plank **100** can be folded or bent at non-right angles by varying the total angle between diagonal surfaces **116**.

(21) The height of vertical surfaces **114** in combination with the width of horizontal surface **112** controls how diagonal surfaces **116** meet when plank **100** is folded. The ideal is to have diagonal surfaces **116** lie flat on each other perfectly aligned so that the entire area of each diagonal surface is contacted by the other diagonal surface. If vertical surfaces **114** are made too short, the top edges of diagonal surfaces **116** will meet first and the diagonal surfaces will not fully touch. If vertical surfaces **114** are made too tall, the bottom edges of diagonal surfaces **116** will meet first and make full contact difficult. The above listed dimensions were found through trial and error to be optimal for most LVP flooring on the market today. However, if diagonal surfaces **116** are not meeting each other properly in practice, some dimensional adjustment might help.

(22) FIG. **3***d* shows plank **100** bent across both grooves **110** to 90-degree angles and glued in place so that the angles are maintained. Adhesive **130** is disposed in grooves **110** prior to folding plank **100**. Diagonal surfaces **116** of each groove **110** contact each other with a thin layer of adhesive **130** between them. Vertical surfaces **114** now form a 90-degree angle, and horizontal surface **112** curves to connect the ends of vertical surfaces **114**.

(23) A gap between horizontal surface **112** and vertical surfaces **114** is shaped like an isosceles right triangle with an outwardly curved hypotenuse. The gap should be filled with adhesive **130** with as few voids as possible to maximize hold of the plank **100** folds. Thorough application of adhesive **130** can be confirmed by viewing a bead formed by the adhesive being squeezed out of groove **110** during folding. If the bead of adhesive **130** is continuous along the length of plank **100** then the gap between horizontal surface **112** and vertical surfaces **114** is likely to be filled with adhesive. Small breaks in the bead of adhesive **130** are likely fine, but long breaks in the bead may indicate an adhesive void in groove **110** at that location.

(24) Bending and gluing both grooves **110** to 90-degree angles completes the transformation of plank **100** into a stair nose **150**. Stair nose **150** is ready to be put into service on a stair step. To install stair nose **150**, glue or adhesive **152** is first applied to bottom surface **102** as shown in FIG. **3***e*. Adhesive **152** can be applied directly to padding **54**. Liquid nails or any type of industrial adhesive can be used. In some embodiments, planks **100** can be manufactured in two different varieties: normal planks with padding **54** for the main floor and planks without padding for stair noses. Both plank varieties are still manufactured together in the same factory to have closely matching color and pattern styles. Leaving padding **54** off planks **100** destined for being made into stair noses **150** has the added benefit that the bead of adhesive **130** that squeezes out of groove **110**

sits on core **52** instead of padding **54**, which provides a stronger adhesive bond.

(25) Adhesive **152** is applied to bottom surface **102** in sufficient quantity to adhere stair nose **150** to the underlying stair tread **12**. In addition, a bead **154** of adhesive **152** is applied over the folded groove **110***b* so that, when stair nose **150** is installed on a stair step as shown in FIG. **3***f*, a gap **160** in the upper corner is substantially filled with adhesive. Filling gap **160** with adhesive **152** structurally supports the corner of stair nose **150** and reduces the likelihood of a heavy stepper breaking or bending the stair nose. For the best structural support, nose **16** of tread **12** should physically contact bottom surface **102** of stair nose **150** between grooves **110***a* and **110***b*. If the bottom section of stair nose **150** is too long such that riser **14** is contacted before nose **16**, then the bottom end of stair nose **150** can be cut off as shown below in FIG. **3***i* to allow nose **16** to be contacted.

(26) With stair nose **150** installed, additional planks **100** can be laid next to the stair nose to continue the rest of the floor as shown in FIGS. **3***g* and **3***h*. Connector **60***a* of plank **100** is interfaced with connector **60***b* of stair nose **150**. As plank **100** is laid down next to stair nose **150**, connectors **60** snap together and create a nearly seamless top surface **104** between the two. For an intermediate stair step, a single plank **100** may be enough to cover the stair tread. Plank **100** may be cut to size so that connector **60***b* is removed and the plank ends at or just short of the riser of the next step. For the top stair step, additional planks **100** are added until the desired floor area is covered. Both plank **100** and stair nose **150** can be cut to length appropriate for the stairs being covered. Using the same planks **100** to form stair nose **150** as well as to cover the surrounding floor area results in a uniform look with consistent color and pattern across the entirety of the floor and stairs.

(27) FIG. **3***i* shows a stair nose **156** made from a plank **100** without padding **54** on the bottom of the plank. Stair nose **156** is otherwise structured and manufactured the same as stair nose **150**. The folded corners of stair nose **156** are stronger than those of stair nose **150** due to adhesive **130** gripping directly to core **52** without the intervening padding **54**. Adhesive **152** between bottom surface **102** and tread **12** functionally and structurally replaces padding **54**. Adhesive **152** fills gap **160** in the corner to structurally support stair nose **156**. The bottom end of stair nose **156** is cut so that nose **16** of tread **12** contacts bottom surface **102** between the two folds. Planks **100** forming the rest of the floor adjacent to stair nose **156** are formed with padding **54**, but still match properly due to being manufactured together at the same factory with the planks used to form the stair nose.

(28) FIGS. **4***a*-**4***c* show an alternative groove profile for converting plank **100** into a stair nose. FIG. **4***a* shows a plan view, FIG. **4***b* shows a cross-sectional view, and FIG. **4***c* shows plank **100** folded into stair nose **192**. Grooves **180** are formed with all square cuts and no diagonal surfaces, which can make manufacturing easier due to the use of blades with perpendicular angles. Grooves **180** include a deep cut **182** to form a flexible portion **184** and a shallow cut **186** to form a shelf **188**. Flexible portion **184** is a thin portion of plank **100** with a uniform thickness across a significant width, similar to flexible portion **120** above. Deep cut **182** and shallow cut **186** can be formed using a single saw blade with an appropriate profile shape or using two or more separate saw blades. Deep cut **182** forms a corner **190** opposite shelf **188**.

(29) Flexible portion **184** is similar to flexible portion **120** in groove **110**, and is formed with a thickness of about ⅔ mm. Some core **52** remains in some embodiments. The width of flexible portion **184** is two to three times greater than the width of flexible portion **120** because the square cut in FIGS. **4***a* and **4***b* will have to cover a large physical distance when plank **100** is folded across groove **180**. Shelf **188** is formed about 1.6 mm deep and 3.2 mm wide. The dimensions of groove **180** can be adjusted as necessary to allow plank **100** to fold properly across the groove.

(30) FIG. **4***c* shows plank **100** folded across grooves **180** to form a stair nose molding **192**. The dimensions of groove **180** are selected so that corner **190** sits on shelf **188** when plank **100** is folded to a 90-degree angle. Groove **180** is filled with adhesive prior to folding, which fills the gap remaining in deep cut **182** after folding. Shallow cut **186** is filled with the plank material from

corner **190**. Groove **180** provides easier manufacturing due to a simpler cut profile and creates a broader radius for the 90-degree bend, which means flooring planks that do not bend as easily can be used. The larger radius bends of grooves **180** may also be a desirable aesthetic choice for some people.

(31) FIGS. **5***a*-**5***e* illustrate saw blades usable to cut grooves **110**. One way to cut groove **110** is to take three normal table saws and shape their teeth to form the three different groove regions, i.e., horizontal surface **112** and the two diagonal surfaces **116**. FIG. **5***a* shows a normal circular saw blade **200** that can be used. Teeth **202** on blade **200** can be shaped as necessary. Buying a blade **200** with as large of teeth **202** as possible will provide the greatest flexibility in shaping the teeth to the desired profile.

(32) FIG. **5***b* shows three saw blades with their teeth cut to make the profile of groove **110**. Middle blade **200***a* has rectangular teeth **202***a*, shaped to the desired width of horizontal surface **112**, e.g., 3.2 mm. Outer blades **200***b* and **200***c* are shaped to have teeth **202***b* and **202***c* with 45-degree outer surfaces to correspond to the desired cuts for diagonal surfaces **116**. Outer blades **200***b* and **200***c* may be made from the exact same circular saw blades as middle blade **200***a*, or a lower diameter blade may be used. Teeth **202** can be shaped using sanding, grinding, or another suitable process.

(33) With blades **200***a*-**200***c* ground down to the desired shapes, the three blades are combined to operate as a single blade on a table saw. FIG. **5***c* illustrates the combined blade **210**. Outer blades **200***b* and **200***c* are rotated slightly toward or away from the viewer so that teeth **202** of the outer blades are interleaved between the teeth of middle blade **200***a*. That rotation allows teeth **202***b* and **202***c* of outer blades **200***b* and **200***c* to extend toward each other into the cut profile of middle blade **200***a* between teeth **202***a*. The angled edges of teeth **202***b* and **202***c* are usually longer than angled surfaces **116** of the resultant grooves **110**.

(34) Combined blade **210** has the appropriate profile to cut groove **110** due to being cut to the proper dimensions. However, the individual blades **200** will eventually need to be sharpened. Keeping the proper saw blade profile after sharpening can be a challenge. The profile of combined blade **210** can be adjusted by adding shims or washers **212** between the individual blades **200***a*-**200***c* as shown in FIG. **5***d*. Moving outer blades **200***b* and **200***c* in the X direction on the illustrated axis adjusts the height in the Y direction where the tops of the outer blades meet middle blade **200***a*. Because the cut angle is 45 degrees, the distance of movement in the X direction will result in an equal distance being added to or removed from vertical surfaces **114** in grooves **110**. Shims **212** allow adjustment of the profile of combined blade **210** to make sure that groove **110** is properly dimensioned.

(35) As an alternative, FIG. **5***e* shows a blade **220** that is a single blade with each individual tooth **222** manufactured in the profile for grooves **110**. Having a single blade **220** means that the profile can no longer be adjusted using shims **212**, but also means that sharpening teeth **222** into the profile of groove **110** is easier. Grooves **110** can also be cut using a router bit with the appropriate profile for cutting grooves **110**. However, using a router bit has the downside of being difficult to sharpen without permanently changing the profile shape.

(36) FIGS. **6***a* and **6***b* illustrate a similar concept for saw blades used to form grooves **180**. Two rectangular blades **200***d* and **200***e* can be combined as shown in FIG. **6***a*. Blade **200***d* has a larger diameter for deep cut **182** and blade **200***e* has a smaller diameter for shallow cut **186**. Again, blades **200***d* and **200***e* can be made from the same input blades, with blade **200***e* simply having more of each tooth removed to reduce the overall diameter and width. Depending on the width of deep cut **182**, two or more saw blades may be combined to form the deep cut while a third makes shallow cut **186**. FIG. **6***b* shows a single blade **230** with each tooth having the profile of groove **180**.

(37) To create grooves using the above blades, the blades are installed into a table saw and planks **100** are run across the table saw. The cutting process begins by optionally heating up planks **100**. A stack or pallet of planks can be placed in a heated area or container prior to having grooves cut. A bread proofing box can be used for instance. Heating planks **100** prior to cutting grooves makes

clearcoat layers **58** more flexible, thus helping reduce the likelihood that the clearcoat layers will chip during the sawing process. Planks **100** are heated to 98 degrees Fahrenheit (° F.) in one embodiment.

(38) FIG. **7**_a_ shows a table saw **240** with a pair of blades **220**_a_ and **220**_b_ disposed on a single axle to cut grooves **110**_a_ and **110**_b_, respectively. Blades **220**_a_ and **220**_b_ are set at a level where the blades cut to the desired depth into plank **100**, i.e., the peak of the blades is 7 and ⅓ mm over the top surface of table saw **240** to create grooves **110** that leave flexible portion **120** with a thickness of ⅔ of a millimeter for 8 mm thick planks. A plank **100** is run across blades **220** using guide **242** to ensure that grooves **110** are positioned properly.

(39) FIGS. **7**_b_ and **7**_c_ show another embodiment where two separate table saws **250**_a_ and **250**_b_ are used to cut grooves **110** one at a time. Cutting one groove at a time with two table saws **250** is a smoother and less error-prone process than doing both grooves at once. Guide **252** keeps planks **100** aligned properly relative to blades **220**. The process of doing two cuts serially can be automated by using motorized rollers **256** to feed a plank **100** into the table saw setup, move the planks from table saw **250**_a_ to table saw **250**_b_, and then drop the plank onto table **260** to await further processing. A second guide **252** can be used on the other side of planks **100** to keep the planks aligned throughout the automated process. Wheels **262** are disposed over blades **220** to keep planks **100** down on the table surface while being cut. Any type of power feeder could be used to move a plank **100** through one or two table saws. A special machine could be made to automatically cut two grooves into plank **100** instead of using two off the shelf table saws.

(40) FIGS. **8**_a_-**8**_c_ illustrates a station **270** used to glue and fold planks **100** after grooves **110** are formed. Station **270** is double-sided so that two planks can be folded and glued at the same time by two different workers standing on opposite sides of the table. Station **270** includes a table **271** with a large flat working surface **272** to support a plank **100**. Alignment pegs **274** are used to align plank **100** parallel to heating slots **276** with the grooves directly over the slots. Plank **100** is set on surface **272** with grooves **110** oriented upward as shown in FIG. **8**_b_, and then slid back against pegs **274**. Two pegs **274** are used to keep the plank **100** grooves parallel to and directly over slots **276**. In other embodiments, more pegs, a flat guide surface, or any other suitable mechanism could be used to keep planks **100** positioned properly on surface **272**. A worker could also just align grooves **110** over slots **276** by sight without an alignment mechanism.

(41) A heating element **280** is disposed under slots **276**. Any type of heating element is usable, e.g., a gas burner or a resistive electric heater. The heating element can be as simple as a food warmer lamp. Slots **276** are positioned directly under grooves **110** with a portion **282** of table **271** limits heat being directly applied to the portion of plank **100** between the grooves. Applying heat specifically to grooves **110** and limiting the application of heat to other areas of planks **100** helps the planks fold at the grooves without bending or being misshapen in other areas. The thinner areas of plank **100** at grooves **110** heat up more quickly than the areas remaining at full thickness, so heating just the grooves is relatively easy. A target temperature of 125° F. is sufficient for folding planks **100** and will keep the planks under most manufacturers' recommended maximum temperature.

(42) Next, adhesive **130** is disposed into grooves **110**. Adhesive **130** is a two-part adhesive in one embodiment. The two-part adhesive involves first spraying an activator into grooves **110** and then dispensing in a bead of glue. Cyanoacrylate (CA) glue is one suitable adhesive. Once the CA glue is applied onto the activator in grooves **110**, the worker has about 10 seconds to fold plank **100** into the desired shape for stair nose **150** before the glue becomes too hard to work.

(43) Another embodiment uses a single-stage hot urethane or polyurethane (PUR) adhesive. The PUR adhesive is dispensed into grooves **110** at a high enough temperature, typically 230° F., that a separate heating element **280** is not required. Using a PUR adhesive to heat the area around grooves **110** provides sufficient heat without needing heating elements **280** and keeps heat localized to the grooves without requiring slots **276**. Adhesive **130** can be dispensed from a bottle,

fed in from a large tank using a hose and nozzle, or applied using any other suitable mechanism.

(44) Once adhesive **130** is disposed in grooves **110**, plank **100** is folded up into two 90-degree angles and placed between table **271** and clamp bar **292** as shown in FIG. **8***c*. Clamp bar **292** runs parallel to the edge of table **271** and is attached to the table by a plurality of flat swing arms **294** that form parallelograms. The top surfaces of swing arms **294** are perpendicular to the inner surfaces of clamp bar **292** and table **271** so that together the three surfaces hold plank **100** folded into two 90-degree angles.

(45) One of the swing arms **294** has a switch **296** extending out past clamp bar **292** that a worker can press with his or her hip to move the clamp bar away from table **271** and allow insertion of a folded-up plank **100**. Clamp bar **292** is spring loaded with spring **297** so that when the worker stops pressing on switch **296** the clamp bar compresses plank **100** between the clamp bar and table **271** to hold the 90-degree folds without additional input from the worker. In other embodiments, springs **297** are used at both ends of clamp bar **292**. FIG. **8***c* shows the spring compression of clamp bar **292** holding the 90-degree angles while the glue dries so that the worker can grab another plank **100** and get heat and glue applied while the first plank's adhesive dries.

(46) The folding of plank **100** will squeeze some adhesive **130** out to form a visible bead inside stair nose **150**. For two-part adhesives, an addition spray of activator can be applied after folding to ensure that the bead hardens. The activator helps adhesive **130** get a better grip on the inside of the folds and reduces the amount that the wet adhesive runs on the inner surfaces of stair nose **150**. 20-30 seconds of drying is typically sufficient for adhesive **130**, and then the completed stair nose **150** can be stacked for packaging and shipment to the customer.

(47) In some embodiments, heating, applying adhesive, folding, and holding while the adhesive dries can all be automated. A robot can apply adhesive before running a plank **100** through a folding machine, such as one that might be used for roll forming sheet metal into channel beams. The entire process from loading a plank **100**, cutting grooves **110** or **180**, to gluing the folds in place can be automated by connecting robots in an assembly line. Robots can be configured to take a pile of new planks **100** and convert the planks into a stack of stair nosings **150** without human intervention.

(48) In addition to stair nosing, other types of molding can be formed by cutting and folding luxury vinyl plank flooring. Any type of molding can be formed, and each has the advantage of matching the surrounding flooring due to being formed from one of the same planks that was used for the flooring.

(49) FIGS. **9***a*-**9***d* show one example where a T molding is made from an LVP flooring plank. FIG. **9***a* shows an LVP strip **300**. LVP strip **300** is formed by cutting plank **100** into strips with the desired length and width for forming a molding. The width W in FIG. **9***a* should be selected greater than the final desired width of the molding in order to accommodate the manufacturing process. In one embodiment, the additional width of strip **300** is between ½ inch and 1 inch.

(50) To form strip **300** into a T molding, the strip is cut or shaved down to the profile shown in FIG. **9***b*. Middle portion **302** stays at the full thickness of plank **100** and operates as the vertical portion of the T molding. Middle portion **302** can be sized as desired for the particular T molding being manufactured. In one embodiment, middle portion **302** has a width suitable for insertion into a metal track that holds the T molding in place. Middle portion **302** can be given sloped side surfaces to apply pressure against track walls as the T molding is inserted.

(51) Platforms **304** surround middle portion **302** on both sides and have bottom surface **102** shaved down to about 20-25% of the total plank **100** thickness, i.e., about 75-80% of the plank material is removed within the footprints of platforms **304**. In one embodiment, a thickness of platforms **304** is about 1 mm and a width of each platform **304** is between ¼ and ½ inch. Platforms **304** will be the portion of the T molding that sits on the surrounding flooring, while middle **302** will be the portion of the T molding that sits between the surrounding flooring.

(52) Flaps **306** have bottom surface **102** of strip **300** shaved down to between ½ mm and ⅔ mm

thickness. The exact thicknesses of flaps **306** and platforms **304** are not critical, but the flaps should be thin enough to be folded under the platforms as shown in FIG. **9***c*. Platforms **304** should be thick enough to allow flaps **306** to be folded under without the platforms being bent.

(53) Platforms **304** and flaps **306** can be cut or shaved down using a single saw with a profile matching the desired shape, as done above for grooves **110** and **180**. One platform **304** and flap **306** could be cut followed by the platform and flap on the other side of middle **302**. Heat can be applied as with grooves **110** and **180** to reduce the likelihood of damaging the clearcoat layers. In another embodiment, a custom planer blade is designed to cut platforms **304** and flaps **306**. Any suitable tool or machine can be used to cut a plank **100** into the shape of FIG. **9***b*. A plank **100** can be cut into a plurality of T-shapes shown in FIG. **9***b* in a single step rather than first cutting down to strips **300**.

(54) Once plank **100** is cut into the shape shown in FIG. **9***b*, flaps **306** are folded under platforms **304** as shown in FIG. **9***c* and glued. Heat can be applied prior to or during folding flaps **306** to reduce the likelihood of image layer **56** and clearcoat layers **58** cracking. A CA, PUR, or other adhesive is used to fix flaps **306** to the undersides of platforms **304**. The width of flaps **306** should be long enough to allow adhesive to sufficiently adhere the flaps to platforms **304** but short enough so that the flaps do not overlap middle **302** when folded under.

(55) Strip **300** with flaps **306** folded under as shown in FIG. **9***c* is usable as a T molding **310**. FIG. **9***d* shows T molding **310** in use. Platform **304** and flap **306** on one side of middle **302** sit on flooring **312** and the other platform and flap sit on flooring **314**. Middle portion **302** extends down between flooring **312** and flooring **314**. T molding **310** can be used anywhere two floorings meet. Flooring **312** might be made of planks **100** while flooring **314** is a tile floor, or the floorings could be two different patterns of LVP planks. T molding **310** can also be used where different areas of the same LVP pattern meet, e.g., if two adjacent rooms were independently covered in the same style of LVP and a molding is needed to cover up a seam between the two.

(56) Whatever the case, T molding **310** covers up the seam where flooring **312** meets flooring **314**. A seamless look by snapping connectors **60** together is difficult to get since the two flooring sides are laid independently. Flooring **312** and **314** are laid with about an inch of space between them, then the gap is covered with T molding **310**. T molding **310** can optionally be glued down or snapped into a track in the gap between floorings **312** and **314**. Because T molding **310** is formed from one of the same planks that are used to make one or both of floorings **312** and **314**, the T molding matches the flooring almost perfectly.

(57) FIGS. **10***a*-**10***e* illustrate forming an end molding from plank **100**. A strip **300** is again cut from plank **100**, and then cut into the profile shown in FIG. **10***a*. Middle portion **302** again remains at the full thickness of planks **100**. One side of middle portion **302** has a platform **304** and a flap **306** as with T molding **310**.

(58) The opposite side of middle **302** has a groove **180** formed to allow that side to fold down at 90 degrees, like the folds done with stair nosing **150**. FIG. **10***b* shows flap **306** folded under platform **304** and glued. Groove **180** is also folded down 90 degrees and glued as when forming stair nosing to complete an end molding **320**. To use end molding **320**, platform **304** and flap **306** are set on flooring as with T molding **310**, and the 90-degree angle on the opposite side extends downward to the underlying floor.

(59) Groove **110** can be used as well as groove **180**. Groove **180** is non-symmetrical and can have shelf **188** disposed toward or away from middle **302**. FIGS. **10***a*-**10***b* form end molding **324** with shelf **188** oriented away from middle **302**, while FIG. **10***c* shows groove **180** cut into strip **300** with shelf **188** oriented toward the middle. FIG. **10***c* also shows an optional extension **322** on the opposite side of groove **180** from middle **302**. Extension **322** creates vertical lift when folded down as shown with end molding **324** in FIG. **10***d*.

(60) The vertical lift of extension **322** allows top surface **104** to stay horizontal when flooring **325** is made from the same thickness of planks **100** as end molding **324**. Extension **322** sits between

two parallel surfaces, i.e., floor **326** and shelf **188**, which helps strengthen end molding **324** from gap **330** being crushed by a person stepping on the end molding. Gap **330** can also be filled with an adhesive or something solid like a strip of plastic, wood, or metal to further strengthen end molding **324**. An additional cut could be made into middle **302** to create a structure sized to be used with a metal track nailed down to the floor.

(61) End molding **320**, with shelf **188** oriented away from middle **302**, could also be made with an extension **322** to lift the grooved side of the end molding. End moldings **320** and **324** are commonly used where LVP flooring ends and a totally different type of flooring is used, e.g., carpet. End moldings **320** and **324** match flooring **325** due to being made from the same planks **100** that the flooring is made from.

(62) The above disclosed methods and devices are described with reference to luxury vinyl plank flooring but apply equally to other types of plank flooring that are sufficiently flexible. For instance, while the illustrated embodiment is made from a luxury vinyl plank (LVP), other type of flooring planks are used in other embodiments. Stone plastic composite (SPC), wood plastic composite (WPC), and engineered vinyl plank (EVP) flooring is a non-exhaustive list of other similar types of flooring that can be used in the above-described method to form molding out of flooring planks.

(63) While two specific groove designs are disclosed, i.e., groove **110** and groove **180**, other groove profiles can be used to allow a floor plank to be bent and used as a molding. Stair noses can be made using any number and angle of folds, e.g., three 60-degree angles could be used instead of two 90-degree angles to create a pointed nose. The total of all fold angles does not necessarily need to equal 180 degrees.

(64) FIGS. **11***a*-**11***g* illustrate an embodiment where a stair nose is made from a flooring plank such that the resulting stair nose is larger than the stair tread being covered. The stair nose is cut to the size of the stair tread prior to installation. FIG. **11***a* shows a plank **400**. Plank **400** is similar to plank **50** but formed with a larger length and width than a typical luxury vinyl floor plank.

(65) Because the stair tread being formed with plank **400** will be used on a step with another vertical riser at the back of the step, plank **400** will not have another plank connected as shown in FIGS. **3***g* and **3***h* above. Therefore, plank **400** is formed without connectors **60** in most embodiments to save the manufacturing work of forming the connectors. Plank **400** is also formed without padding **54** because the padding is not necessary to match the thickness of an adjacent plank. Adhesive used to stick the plank to a stair tread will provide sufficient padding. Forming plank **400** without padding also strengthens the glued corner joints as described above.

(66) Plank **400** is a plank formed specifically for making a stair nose, with specific characteristics that are adapted to that purpose, i.e., greater size, no padding **54**, and no connectors **60**. Plank **400** is still formed with the same core **52**, image layer **56**, and clearcoat **58** as other floor planks **50** which are used on the surrounding floors to ensure that the look and feel of the stair noses matches the floors.

(67) In FIG. **11***b*, grooves **110** are formed along the length of plank **400** as described above. Grooves **110** are formed with a uniform cross-section along the entire length of the plank **400** as described above and are spaced out and positioned similarly. The specific positions of grooves **110** can be adjusted as needed to make a different size stair nose. Square grooves like grooves **180**, or any other suitable shape of grooves, are used in other embodiments.

(68) FIGS. **11***c* and **11***d* show plank **400** folded and glued into a stair nose **420**. Short edges **108** and long edges **106** are flat rather than having connectors **60**. Adhesive **130** is disposed into grooves **110** prior to folding to hold the fold at approximately a 90-degree angle.

(69) FIG. **11***e* shows a set of stairs **430**. Each tread **12** has a length L and a width W that is less than the comparable dimensions of stair nose **420**. Therefore, stair nose **420** can be cut down to the length and width of tread **12** and then a single piece of molding can be used to entirely finish each tread. There is no need to connect multiple stair noses together to cover the length of a tread **12** or

to connect additional pieces of flooring to fill width of the tread.

(70) FIG. **11***f* shows two cuts **436** and **438** that can be made to stair nose **420** to give the stair nose the same length L and width W as tread **12**. Cut **436** is parallel to long edges **106** to reduce the width of stair nose **420** down to width W of tread **12**. Cut **438** is parallel to short edge **108** to reduce the length of stair nose **420** down to length L. Cuts **436** and **438** can be performed using a table saw, circular saw, jig saw, hand saw, laser cutting tool, water jet, or any other suitable cutting mechanism. In other embodiments, stair nose **420** is manufactured to length L and width W, so no cuts are necessary to have a plank the same size as a stair step.

(71) FIG. **11***g* shows stair noses **420** being installed on stairs **430**. Installation typically proceeds from bottom of the stairs to the top. Stair nose **420***a* is installed on tread **12***a* using adhesive **152**. Back edge **106***a* of nose **420***a* is oriented toward the next higher riser **14**.

(72) The length and width of the installed stair nose **420** may intentionally be formed to not exactly match the length and width of a stair nose **12***a* for a variety of reasons. As one example, the width of stair nose **420** may be a millimeter or two short to ensure a proper fit on all steps considering potential variances in the exact width between steps. A small gap will therefore be left between riser **14** and edge **106***a*. A riser cover **440***a* is installed over riser **14** with adhesive **152** and covers up any small gap between nose **420***a* and the riser. Riser cover **440***a* is formed from the same material as stair noses **420** to match the aesthetics of the surrounding floor and stairs. In other embodiments, riser covers **440** that visually contrast with stair noses **420** are used.

(73) Stair nose **420***b* is installed on the next tread **12***b*. Edge **106***b* of stair nose **420***b* can touch riser cover **440***a* or a gap may be left. In other embodiments, riser cover **440***a* is shorter than riser **14**, and edge **106***b* may be directly over or behind the riser cover. Edge **106***b* touches riser **14** in some embodiments. Each stair nose **420** fully covers a tread **12** without needing additional floor panels to fill up a gap between the edge **106***a* and riser **14**.

(74) For longer steps, two or more stair noses **420** can still be used to fully cover a tread **12**, but each nose **420** would typically still be wide enough to reach from nose **16** to the next riser **14**. In such cases, connectors **60** can be formed on shorter edges **108**. Connectors **60** would be cut off at the ends of each stair tread **12** but would be used internally to connect multiple stair noses **420** to reach the desired length. Connector **60** is cut off or never formed on longer edges **106**. Connector **60** can be formed on top edge **106***a* only so that stair noses **420** can be linked up with surrounding flooring at the top of stairs **430**.

(75) Using stair noses **420** that are made over-sized for the tread being covered and then cutting the stair noses to size prior to installation simplifies installation and improves aesthetics by reducing the number of seams between panels and the total number of panels that must be placed. Any of the above or below embodiments can be formed from an oversized piece of flooring and then cut to the desired size.

(76) FIGS. **12***a*-**12***h* show an embodiment where stair noses are made from thinner and cheaper vinyl flooring. FIG. **12***a* shows a panel of vinyl flooring **450**. Vinyl flooring **450** is significantly thinner and more flexible than the luxury plank flooring used above. Vinyl flooring **450** is typically installed by gluing a plurality of pieces of the vinyl flooring onto the floor directly abutting each other. There is no latching system as luxury vinyl planks typically have. The term vinyl may refer to PVC or another vinyl-based or vinyl-like material.

(77) Vinyl flooring **450** is an economical option, but value would still be added by having stair nosing that matched the surrounding flooring. However, due to vinyl flooring **450** being thin and flexible, performing the above-described groove cutting, folding, and gluing will not make a satisfactory stair nose when performed on the vinyl flooring by itself.

(78) FIG. **12***b* shows a backing **452** that can be attached to vinyl flooring **450** to provide enough thickness and strength to form a stair nose. Backing **452** is formed from PVC or vinyl similarly to vinyl flooring **450**. Backing **452** is thick enough to allow grooves **110** or **180** to be formed when combined with the thickness of vinyl flooring **450**. An adhesive **454** is applied to one surface of

backing **452** using a spray can, a spray nozzle as part of an automated system, a brush, or another suitable means.

(79) FIG. **12***c* shows backing **452** attached onto vinyl flooring **450**. The surface of backing **452** to which adhesive **454** was applied is oriented toward vinyl flooring **450**. In one embodiment, adhesive **454** is a type of adhesive or PVC cement that eats or melts the vinyl material of flooring **450** and backing **452** and then resolidifies to essentially weld the two pieces together. Any suitable adhesive is used in other embodiments. Backing **452** has a smaller width than vinyl flooring **450** so that only the portion of the vinyl flooring that will be grooved and folded has backing **452**.

(80) FIG. **12***d* shows a cross-section of vinyl flooring **450** with backing **452** attached. Vinyl flooring **450** typically has a vinyl substrate **460**, an image layer, and clearcoat layer **462** applied over the image layer, similar to image layer **56** and clearcoat layer **58** above. In other embodiments, vinyl flooring is used that is just a vinyl substrate with a visual pattern embedded in or painted onto the vinyl material.

(81) In FIG. **12***e*, beveled grooves **110***a* and **110***b* are formed as described above. Square grooves **180** or any other suitable groove profile is formed in other embodiments. Grooves **110** extend along the entire length of vinyl flooring **450** and backing **452**. Grooves **110** are formed extending completely through backing **452** and into vinyl flooring **450** to a depth that is just to the image layer without extending through to clearcoat layer **462**. A portion of vinyl substrate **460** remains at the bottoms of grooves **110** in other embodiments. Grooves **110** can also be formed only in backing **452** and not extending into vinyl flooring **450** in embodiments with especially thin vinyl flooring or other types of thin flooring. FIG. **12***f* shows vinyl flooring **450** and backing **452** from the bottom with grooves **110**.

(82) The portion of vinyl flooring **450** that has backing **452** attached is folded around grooves **110** and glued as described above to form a stair nose **470** in FIG. **12***g*. Stair nose **470** is attached to tread **12** by adhesive **152** or another suitable adhesive. A size of backing **462** is selected so that edge **472** running parallel to nose **16** does not contact tread **12** or nose **16**. Therefore, vinyl flooring **450** of stair nose **470** lies flat on tread **12** like the next piece of flooring **450***a* that is not part of a stair nose. In some embodiments, backing **452** is cut short enough that edge **472** is coplanar with the underlying vertical section of the backing. Because only flooring **450** without backing **452** is placed on tread **12**, the thickness and feel of the stair nose matches the rest of the surrounding vinyl flooring.

(83) An edge of backing **452** does not necessarily need to align perfectly with an edge of vinyl flooring **450**. FIG. **12***h* shows an embodiment where edge **474** opposite edge **472** is located inward from edge **106***b* of flooring **450**. The portion of vinyl flooring **450** under tread **12** is not going to need to support significant weight during normal usage, so having the edge of the vinyl flooring floating without backing **452** is not going to be structurally problematic. In other embodiments, backing **452** is split into a separate piece per groove **110**, such that each corner **480***a* and **480***b* has a separate piece of backing material.

(84) While one or more embodiments of the present invention have been illustrated in detail, the skilled artisan will appreciate that modifications and adaptations to those embodiments may be made without departing from the scope of the present invention as set forth in the following claims.

## Claims

1. A flooring system stair nose molding, comprising: a vinyl flooring; a vinyl backing attached to a bottom surface of the vinyl flooring, wherein the vinyl backing extends to a first edge of the vinyl flooring and a second edge of the vinyl flooring opposite the first edge such that a length of the vinyl backing is approximately equal to a length of the vinyl flooring, and wherein the vinyl flooring includes a third edge outside a footprint of the vinyl backing such that a first portion of the bottom surface of the vinyl flooring is exposed between the third edge and the vinyl backing; a first

groove formed through the vinyl backing and into the vinyl flooring from the first edge to the second edge; and a second groove formed through the vinyl backing and into the vinyl flooring from the first edge to the second edge in parallel with the first groove; wherein the first groove and second groove define, a first section of the vinyl flooring adjacent to the first groove opposite the second groove, a second section of the vinyl flooring between the first groove and second groove, and a third section of the vinyl flooring adjacent to the second groove opposite the first groove; wherein the first section is larger than the third section; wherein the first portion of the bottom surface is part of the first section; wherein the vinyl flooring is folded at the first groove and second groove; and a stair tread including a flat top surface and a rounded nose, wherein the vinyl flooring is directly attached to the flat top surface of the stair tread with an adhesive, wherein the adhesive extends from the stair tread to the first portion of the bottom surface of the vinyl flooring, and wherein an edge of the vinyl backing in the first section and oriented away from the second section is laterally offset from the flat top surface of the stair tread.

2. The stair nose molding of claim 1, wherein the vinyl backing is attached to the vinyl flooring with a plastic cement.

3. The stair nose molding of claim 1, wherein the vinyl flooring is folded at a 90-degree angle at both the first groove and second groove.

4. The stair nose molding of claim 1, further including an adhesive disposed in the first groove.

5. The flooring system of claim 1, wherein the vinyl flooring includes a fourth edge, opposite the third edge, that is outside a footprint of the vinyl backing such that a second portion of the vinyl flooring is exposed between the vinyl backing and the fourth edge of the vinyl flooring.

6. The flooring system of claim 1, wherein an edge of the vinyl backing in the first section of the vinyl flooring is coplanar to a surface of the vinyl backing in the section of the vinyl flooring.

7. A molding, comprising: a piece of flooring; a backing attached to a first portion of a bottom surface of the piece of flooring, wherein a second portion of the bottom surface remains exposed from the backing; a first groove formed in the backing; and a second groove formed in the backing, wherein the piece of flooring and backing are folded at the first groove and second groove, wherein a size of a first section of the piece of flooring on a first side of the first groove opposite the second groove is greater than a size of a second section of the piece of flooring on a second side of the second groove opposite the first groove, and wherein the exposed second portion of the bottom surface is part of the first section of the piece of flooring, wherein a width of the backing extending from the first groove to the second groove is greater than a width of the backing in the first section of the piece of flooring in a direction perpendicular to the first groove.

8. The molding of claim 7, wherein the piece of flooring and backing both comprise vinyl or PVC.

9. The molding of claim 7, further including an adhesive disposed in the first groove and second groove.

10. The molding of claim 7, wherein the second portion of the bottom surface comprises a majority of the first section of the piece of flooring in plan view.

11. A flooring system comprising the molding of claim 7, further including a stair tread, wherein the molding is disposed over the stair tread with the second portion of the bottom surface attached to the stair tread with an adhesive.

12. The flooring system of claim 11, wherein an edge of the backing in the first section of the piece of flooring is laterally offset from a top surface of the stair tread.

13. A method of making a molding, comprising: providing a piece of flooring; attaching a backing to a bottom surface of the piece of flooring, wherein a portion of the bottom surface of the piece of flooring remains exposed from the backing; forming a first groove through the backing and into the piece of flooring after attaching the backing to the bottom surface of the piece of flooring; forming a second groove through the backing and into the piece of flooring after attaching the backing to the bottom surface of the piece of flooring, wherein a first section of the piece of flooring on a first side of the first groove opposite the second groove is larger than a second section of the piece of

flooring on a second side of the second groove opposite the first groove, and wherein the exposed portion of the bottom surface is part of the first section; and folding the piece of flooring at the first groove and second groove.

14. The method of claim 13, further including attaching the backing to the piece of flooring using a plastic cement.

15. The method of claim 13, further including forming the first groove and second groove completely through the backing and into the piece of flooring.

16. The method of claim 13, wherein the exposed portion of the bottom surface of the piece of flooring comprises a majority of the first section of the piece of flooring in plan view.

17. The method of claim 13, wherein a second portion of the bottom surface of the piece of flooring is exposed from the backing after attaching the backing to the bottom surface of the piece of flooring, and wherein the second portion of the bottom surface is part of the second section after forming the first groove and second groove.

18. A flooring method comprising: making the molding according to the method of claim 13; and disposing the molding on a stair tread.

19. The method of claim 18, further including attaching the molding to the stair tread with the backing physically separated from the stair tread.

20. The method of claim 18, wherein an edge of the backing in the first section of the flooring is laterally offset from a top surface of the stair tread.